

MULTILEVEL INHERITANCE

Date: 24/07/2026

Objective / Aim

To demonstrate inheritance, method overriding, method overloading, and the use of inherited members in Java.

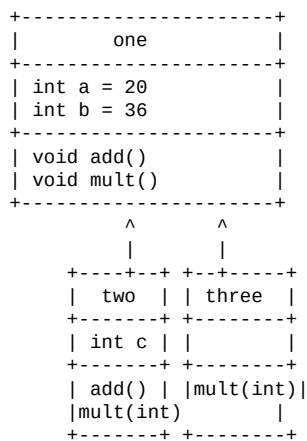
Algorithm / Logic

1. Create the base class one with variables a and b.
2. Define add() and mult() methods in class one.
3. Create class two extending one and add variable c.
4. Override add() in class two and overload mult() with an integer parameter.
5. Create class three extending one and define another overloaded mult() method.
6. Create an object of class two in main().
7. Call mult(5) and the inherited mult() method.

Steps for Execution

1. Create the Java project and add the Multilevel class.
2. Compile and run Multilevel.java.
3. Create an object of class two.
4. Call the overloaded mult(int) method.
5. Call the inherited mult() method.
6. Observe the output.

UML Class Diagram



Source Code

```
package inheritance;

class one{
    int a = 20;
    int b = 36;

    void add() {
        System.out.println("sum of the numbers:" + (a+b+100) );
    }

    void mult() {
        System.out.println("mult"+ (a*2) );
    }
}

class two extends one{
    int c = 30;

    void add() {
        System.out.println("sum of the numbers"+(a+b+c));
    }

    void mult(int e) {
        System.out.println("mult"+ (e*50) );
    }
}

class three extends one {
    void mult(int d) {
```

```
        System.out.println("mult"+ (a*b*2));
    }
}

public class Multilevel {
    public static void main(String[] args) {
        two kc = new two();
        kc.mult(5);
        kc.mult();
    }
}
```

Output

```
mult250
mult40
```

SINGLE INHERITANCE

Date: 24/07/2026

Objective / Aim

To demonstrate single inheritance, method overriding, and method overloading in Java.

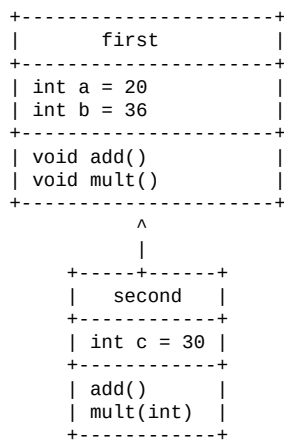
Algorithm / Logic

1. Create the base class first with variables a and b.
2. Define add() and mult() methods in class first.
3. Create class second extending first and add variable c.
4. Override add() in class second.
5. Overload mult() by defining mult(int e) in class second.
6. Create an object of class second in main().
7. Call add(), mult(35), and the inherited mult() method.

Steps for Execution

1. Create the Java project and add the Single class.
2. Compile and run Single.java.
3. Create an object of class second.
4. Call the overridden add() method.
5. Call the overloaded mult(int) method.
6. Call the inherited mult() method and observe the output.

UML Class Diagram



Source Code

```
package inheritance;

class first {
    int a = 20;
    int b = 36;

    void add() {
        System.out.println("sum of the numbers:" + (a+b+100) );
    }

    void mult() {
        System.out.println("mult"+ (a*2) );
    }
}

class second extends first{
    int c = 30;

    @Override
    void add() {
        System.out.println("sum of the numbers" +(a+b+c));
    }

    //overload
    void mult(int e) {
        System.out.println("mult"+ (e*50) );
    }
}

public class Single {
```

```
        public static void main(String[] args) {  
            second sc = new second();  
            sc.add();  
            sc.mult(35);  
            sc.mult();  
        }  
    }
```

Output

```
sum of the numbers86  
mult1750  
mult40
```

SUPER KEYWORD

Date: 24/07/2026

Objective / Aim

To demonstrate method overriding and the use of the super keyword to call superclass methods in Java.

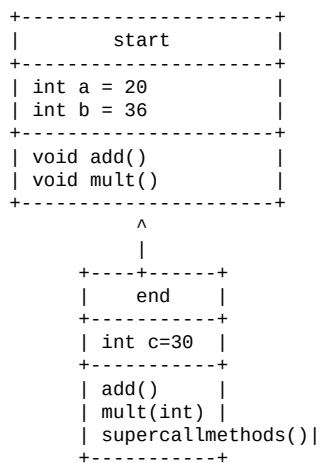
Algorithm / Logic

1. Create the superclass start with variables a and b.
2. Define add() and mult() methods in class start.
3. Create class end extending start and add variable c.
4. Override add() in class end.
5. Define an overloaded mult(int e) method in class end.
6. Define supercallmethods() using super.add() and super.mult().
7. Create an object of class end and call the required methods.

Steps for Execution

1. Create the Java project and add the Super class.
2. Compile and run Super.java.
3. Create an object of class end.
4. Call supercallmethods() to execute superclass methods.
5. Call the overridden add() method.
6. Call mult(35) and observe the output.

UML Class Diagram



Source Code

```
package inheritance;

class start{
    int a = 20;
    int b = 36;

    void add() {
        System.out.println("sum of the numbers:" + (a+b+100) );
    }

    void mult() {
        System.out.println("mult"+ (a*2) );
    }
}

class end extends start{
    int c = 30;

    void add() {
        System.out.println("sum of the numbers" +(a+b+c));
    }

    void mult(int e) {
        System.out.println("mult"+ (e*50) );
    }

    void supercallmethods() {
        super.add();
    }
}
```

```
        super.mult();
    }
}

public class Super {
    public static void main(String[] args) {
        end ex = new end();
        ex.supercallmethods();
        ex.add();
        ex.mult(35);
    }
}
```

Output

```
sum of the numbers:156
mult40
sum of the numbers86
mult1750
```